

Project Title:
Statistical Analysis of Student Academic Performance

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Abstract

This report presents a master's-level statistical investigation of student academic performance using a publicly available dataset analyzed in SAS (N = 995). The study applies descriptive statistics, Pearson correlation, an independent two-sample t-test, chi-square test of independence, one-way ANOVA with Tukey HSD, and binary logistic regression. Across the six research questions, descriptive results indicate moderate average academic performance, while inferential findings provide limited statistical evidence that the selected predictors explain performance differences or pass/fail outcomes.

Introduction

Student performance research often evaluates whether behavioral variables such as study habits, attendance, and sleep are associated with academic outcomes. Study behaviors and academic attitudes have been linked to performance in prior work (Credé & Kuncel, 2008), and sleep has been discussed as a factor related to learning capacity (Curcio et al., 2006). Using a public educational dataset (Poonia, 2022) and SAS OnDemand for Academics for analysis, this project tests six research questions aligned with CCDS620 methods.

Project Goal

To apply statistical techniques to describe student performance and test whether selected demographic and behavioral variables are associated with academic outcomes.

Problem / Decision-Making Statement

Which student-related factors are associated with academic outcomes, and can selected behavioral variables predict pass/fail results?

Research Questions

- **RQ1:** What is the distribution of students' average academic scores?
- **RQ2:** Is there a statistically significant relationship between mathematics scores and science scores?
- **RQ3:** Is there a significant difference in sleep hours between students who passed and students who failed?
- **RQ4:** Is there a significant association between gender and final academic result?
- **RQ5:** Do average academic scores differ significantly across attendance levels?
- **RQ6:** Can pass/fail outcomes be predicted using study hours, attendance, sleep hours, and internet usage?

Dataset and Data Preparation

The analysis dataset (WORK.STUDENT_ANALYSIS) contains 995 observations and 12 variables, including three derived variables (Average_Score, Pass_Flag, and Attendance_Level). Attendance_Level appears truncated in the SAS output (Hig/Med/Low) due to character length, but it represents High/Medium/Low categories. Figures 1–5 summarize dataset structure and key categorical distributions.

```
/* Derived variables created for analysis */  
data WORK.STUDENT_ANALYSIS;  
  set WORK.STUDENT_RAW;  
  
  Average_Score = mean(Math_Score, Science_Score);  
  
  if Final_Result = 'Pass' then Pass_Flag = 1;  
  else if Final_Result = 'Fail' then Pass_Flag = 0;  
  
  length Attendance_Level $ 6;  
  if Attendance < 70 then Attendance_Level = 'Low';  
  else if Attendance < 85 then Attendance_Level = 'Med';  
  else Attendance_Level = 'Hig';  
run;
```

Attendance_Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Hig	306	30.75	306	30.75
Low	377	37.89	683	68.64
Med	312	31.36	995	100.00

Figure 1 Attendance_Level

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
4	Attendance	Num	8	BEST12.	BEST32.
12	Attendance_Level	Char	10		
10	Average_Score	Num	8		
9	Final_Result	Char	4	\$4.	\$4.
2	Gender	Char	1	\$1.	\$1.
8	Internet_Usage_Hours	Num	8	BEST12.	BEST32.
5	Math_Score	Num	8	BEST12.	BEST32.
11	Pass_Flag	Num	8		
6	Science_Score	Num	8	BEST12.	BEST32.
7	Sleep_Hours	Num	8	BEST12.	BEST32.
1	Student_ID	Num	8	BEST12.	BEST32.
3	Study_Hours	Num	8	BEST12.	BEST32.

Figure 2. Variables and attributes in WORK.STUDENT_ANALYSIS.

Descriptive Statistical Summary

Measure	Value
N	995
Mean (Average_Score)	65.31098
Median (Average_Score)	65.00000
Std Deviation (Average_Score)	14.00784
Min–Max (Average_Score)	31.0 – 97.5
Q1–Q3 (Average_Score)	54.5 – 75.675

To enhance interpretability, Figures 6–8 provide simple visual summaries of key categorical distributions derived from the SAS frequency outputs.

RQ1: Distribution of Students' Average Academic Scores

Method

Descriptive statistics and distributional plots were used to summarize Average_Score.

SAS Code Used

```
proc means data=WORK.STUDENT_ANALYSIS n mean median std min max q1 q3;
    var Average_Score;
run;

proc univariate data=WORK.STUDENT_ANALYSIS;
    var Average_Score;
    histogram Average_Score / normal;
run;
```

Key Numerical Results

Measure	Value
Mean	65.31098
Median	65.00000
Std Deviation	14.00784
Q1	54.5
Q3	75.675
Min	31.0
Max	97.5
Decision at $\alpha = 0.05$	Descriptive (no hypothesis test)

SAS Output Evidence and Illustrations

Basic Statistical Measures			
Location		Variability	
Mean	65.31098	Std Deviation	14.00784
Median	65.00000	Variance	196.21966
Mode	71.50000	Range	66.50000
		Interquartile Range	21.17500

Figure 3. Basic measures of location and variability (UNIVARIATE).

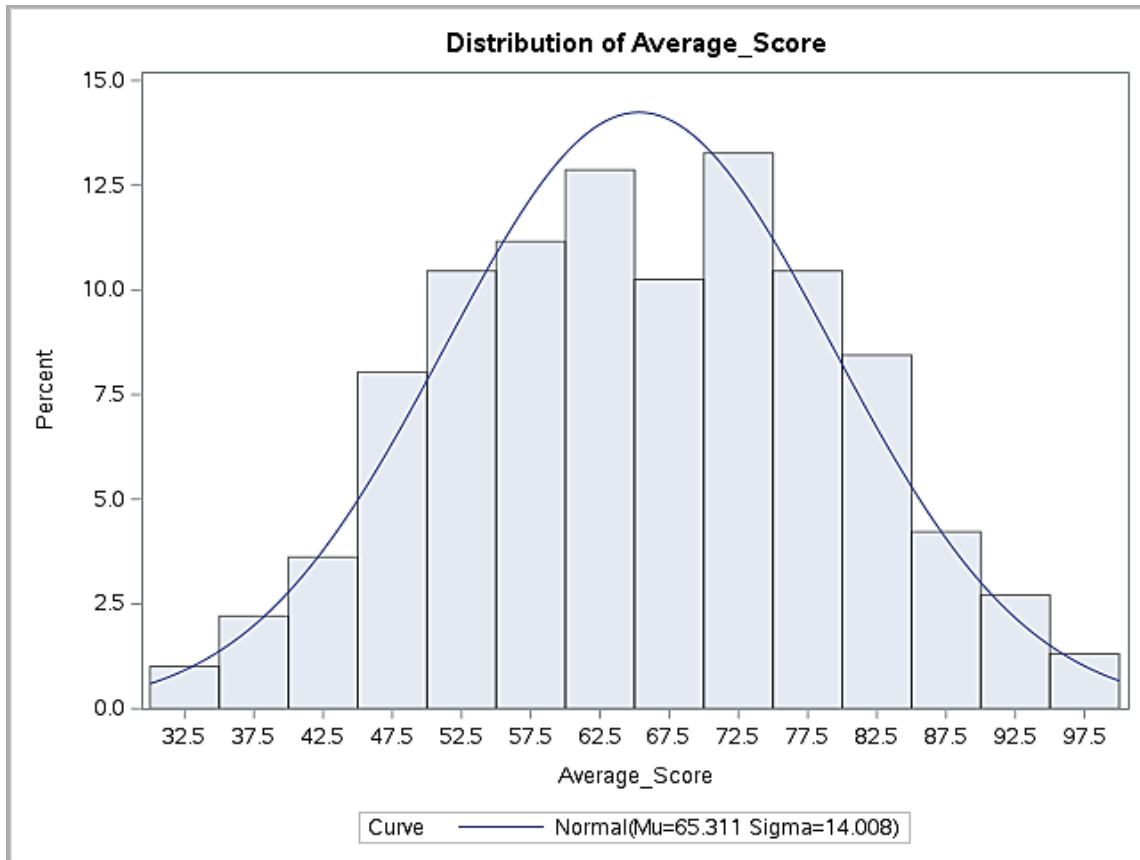


Figure 4. Histogram for Average_Score.

Interpretation

Average_Score is centered around the mid-60s (Mean ≈ 65.31) with moderate dispersion (SD ≈ 14.01). Normality tests in the UNIVARIATE output indicate deviations from a normal distribution, which is common in educational score data.

RQ2: Relationship Between Mathematics Scores and Science Scores

Hypotheses

H0: There is no linear relationship between Math_Score and Science_Score.

H1: There is a linear relationship between Math_Score and Science_Score.

Method

Pearson correlation analysis assessed the linear association between Math_Score and Science_Score.

SAS Code Used

```
proc corr data=WORK.STUDENT_ANALYSIS pearson;  
    var Math_Score Science_Score;  
run;  
  
proc sgplot data=WORK.STUDENT_ANALYSIS;  
    scatter x=Math_Score y=Science_Score;  
    reg x=Math_Score y=Science_Score;  
run;
```

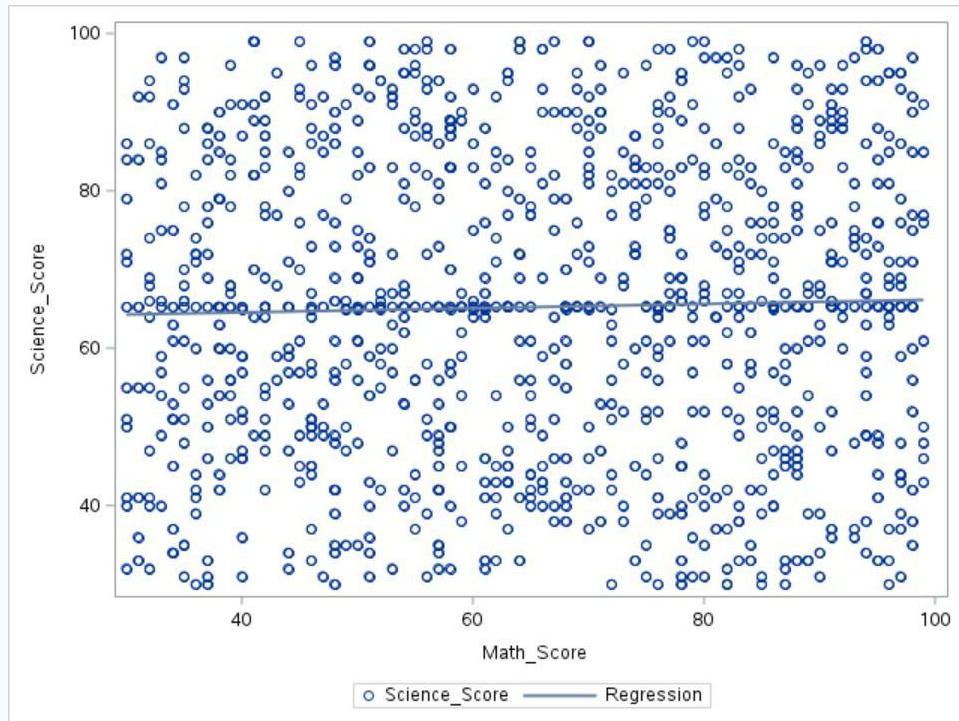
Key Numerical Results

Measure	Value
Pearson r	0.02842
p-value	0.3704
N	995
Decision at $\alpha = 0.05$	Fail to Reject H0

SAS Output Evidence and Illustrations

Pearson Correlation Coefficients, N = 995 Prob > r under H0: Rho=0		
	Math_Score	Science_Score
Math_Score	1.00000	0.02842 0.3704
Science_Score	0.02842 0.3704	1.00000

Figure 5. Pearson correlation coefficients table.



+

Figure 6. Scatter plot of Math_Score vs Science_Score.

Interpretation

The correlation is near zero and not statistically significant ($p = 0.3704$), providing insufficient evidence of a meaningful linear relationship between mathematics and science scores.

RQ3: Difference in Sleep Hours Between Passed and Failed Students

Hypotheses

H0: Mean Sleep_Hours is equal for pass and fail groups.

H1: Mean Sleep_Hours differs between pass and fail groups.

Method

An independent two-sample t-test compared Sleep_Hours between Final_Result groups.

SAS Code Used

```
proc ttest data=WORK.STUDENT_ANALYSIS;
  class Final_Result;
  var Sleep_Hours;
run;
```

Key Numerical Results

Measure	Value
Fail mean Sleep_Hours	6.9528
Pass mean Sleep_Hours	6.9827
t (pooled)	-0.43
p-value	0.6709
Decision at $\alpha = 0.05$	Fail to Reject H0

SAS Output Evidence and Illustrations

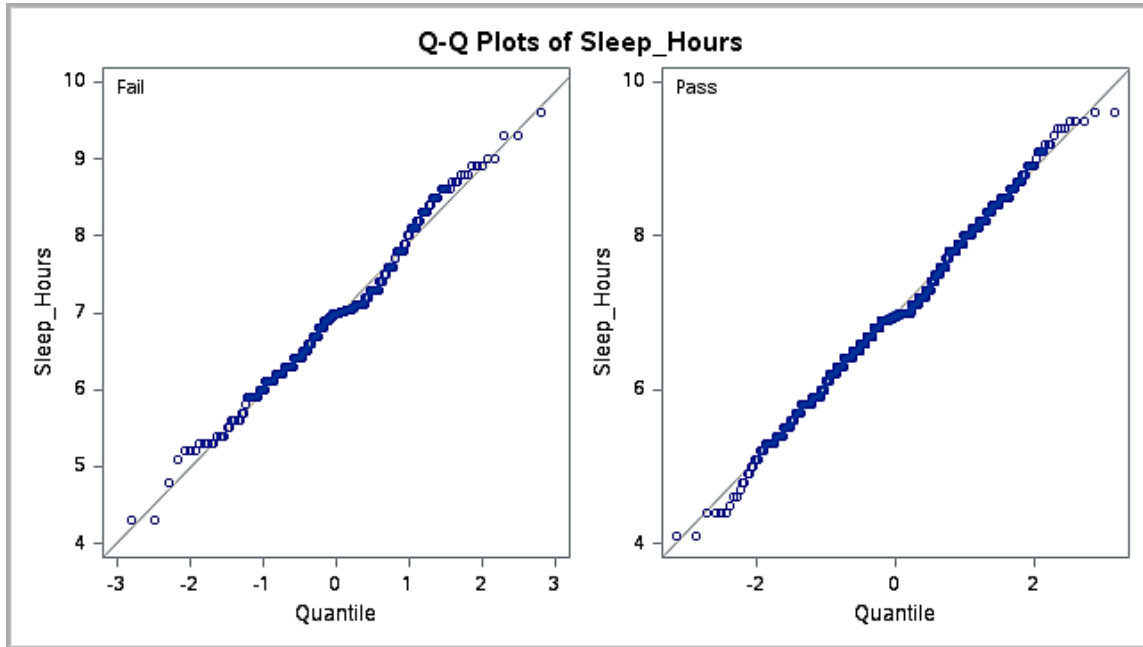


Figure 7. Q-Q plot for Sleep_Hours (distributional check).

Interpretation

Sleep_Hours does not differ significantly between pass and fail groups ($p = 0.6709$). The mean difference is small and the confidence interval includes zero.

RQ4: Association Between Gender and Final Academic Result

Hypotheses

H0: Gender and Final_Result are independent.

H1: Gender and Final_Result are associated.

Method

A chi-square test of independence evaluated the association between Gender and Final_Result.

SAS Code Used

```
proc freq data=WORK.STUDENT_ANALYSIS;  
  tables Gender*Final_Result / chisq measures;  
run;
```

Key Numerical Results

Measure	Value
Chi-square	0.8970
df	1
p-value	0.3436
Cramer's V	0.0300
Decision at $\alpha = 0.05$	Fail to Reject H0

SAS Output Evidence and Illustrations

Statistics for Table of Gender by Final_Result

Statistic	DF	Value	Prob
Chi-Square	1	0.8970	0.3436
Likelihood Ratio Chi-Square	1	0.8978	0.3434
Continuity Adj. Chi-Square	1	0.7629	0.3824
Mantel-Haenszel Chi-Square	1	0.8961	0.3438
Phi Coefficient		0.0300	
Contingency Coefficient		0.0300	
Cramer's V		0.0300	

Figure 8. Contingency table ($\text{Gender} \times \text{Final_Result}$).

Interpretation

The chi-square test is not significant ($p = 0.3436$), and Cramer's V indicates a negligible association (≈ 0.03). The pass-rate chart shows only minor differences between categories.

RQ5: Differences in Average Academic Scores Across Attendance Levels

Hypotheses

H0: Mean Average_Score is equal across attendance levels.

H1: At least one attendance level has a different mean Average_Score.

Method

One-way ANOVA tested differences in Average_Score across Attendance_Level groups; Tukey HSD examined pairwise differences.

SAS Code Used

```
proc anova data=WORK.STUDENT_ANALYSIS;  
  class Attendance_Level;  
  model Average_Score = Attendance_Level;  
  means Attendance_Level / tukey;  
run;quit;
```

Key Numerical Results

Measure	Value
F statistic	1.47
p-value	0.2315
R-Square	0.002945
Decision at $\alpha = 0.05$	Fail to Reject H0

SAS Output Evidence and Illustrations

Dependent Variable: Average_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	574.4622	287.2311	1.47	0.2315
Error	992	194467.8831	196.0362		
Corrected Total	994	195042.3453			

Figure 9. Dependent Variable.

Interpretation

Average_Score does not differ significantly by attendance level ($p = 0.2315$). Tukey HSD indicates no significant pairwise differences (all confidence intervals include zero).

RQ6: Predicting Pass/Fail Outcomes Using Logistic Regression

Hypotheses

H0: Study_Hours, Attendance, Sleep_Hours, and Internet_Usage_Hours do not predict Pass_Flag.

H1: At least one predictor significantly predicts Pass_Flag.

Method

Binary logistic regression modeled the probability of passing (Pass_Flag = 1) using four predictors.

SAS Code Used

```
proc logistic data=WORK.STUDENT_ANALYSIS;  
    model Pass_Flag(event='1') = Study_Hours Attendance Sleep_Hours  
    Internet_Usage_Hours;  
run;
```

Key Numerical Results

Measure	Value
Likelihood Ratio χ^2	1.7362
df	4
p-value	0.7841
c-statistic	0.523
Decision at $\alpha = 0.05$	Fail to Reject H0

SAS Output Evidence and Illustrations

Response Profile		
Ordered Value	Pass_Flag	Total Frequency
1	0	244
2	1	751

Probability modeled is Pass_Flag=1.

Figure 10. Response profile (Pass_Flag).

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1110.489	1116.752
SC	1115.391	1141.266
-2 Log L	1108.489	1106.752

Figure 11. Model fit statistics.

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	1.7362	4	0.7841
Score	1.7352	4	0.7843
Wald	1.7322	4	0.7849

Figure 12. Global tests (BETA=0).

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	0.9789	0.7196	1.8502	0.1738
Study_Hours	1	0.0563	0.0664	0.7197	0.3963
Attendance	1	-0.00200	0.00514	0.1514	0.6972
Sleep_Hours	1	0.0346	0.0773	0.2011	0.6539
Internet_Usage_Hours	1	-0.0234	0.0287	0.6662	0.4144

Figure 13. Parameter estimates.

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	52.3	Somers' D	0.047
Percent Discordant	47.7	Gamma	0.047
Percent Tied	0.0	Tau-a	0.017
Pairs	183244	c	0.523

Figure 14. Predict Probabilities.

Interpretation

The logistic model is not statistically significant ($p = 0.7841$) and shows weak discrimination ($c = 0.523$). None of the predictors have statistically significant coefficients, and odds ratios are close to 1, indicating minimal practical impact in this dataset.

Integrated Summary of Findings

RQ	Technique	Key Result	Decision
RQ1	Descriptive	Mean=65.31, SD=14.01	—
RQ2	Pearson r	$r=0.028$, $p=0.3704$	Fail to Reject H_0
RQ3	t-test	$p=0.6709$	Fail to Reject H_0
RQ4	Chi-square	$p=0.3436$, $V=0.030$	Fail to Reject H_0
RQ5	ANOVA/Tukey	$p=0.2315$	Fail to Reject H_0
RQ6	Logistic regression	Model $p=0.7841$, $c=0.523$	Fail to Reject H_0

Overall, inferential analyses consistently indicate weak or non-significant relationships between the selected variables and academic outcomes in this dataset. The results highlight the need for additional predictors (e.g., prior achievement, motivation, instructional context) to improve explanatory and predictive power.

Conclusion

This project demonstrates correct selection and application of statistical methods in SAS for educational data analysis. While most tests were not statistically significant, these findings are substantively informative because they indicate that the included predictors explain little variation in student outcomes within this dataset.

Limitations

The dataset includes a limited set of predictors and may not capture broader socioeconomic, psychological, or instructional factors. Additionally, derived categorical labels in the SAS output (e.g., Hig/Med/Low) are abbreviated due to character length constraints.

Recommendations

Future analysis should incorporate additional variables (e.g., previous GPA, assessment components, motivation indicators) and consider alternative models (e.g., non-linear effects, interaction terms) to improve predictive performance.

References

Poonia, S. (2022). Educational purpose dataset [Data set]. Kaggle.
<https://www.kaggle.com/datasets/sumitpoonia/educational-purpose-dataset/data?select=Result.csv>

SAS Institute Inc. (n.d.). SAS® OnDemand for Academics: Uploading and Accessing Local Data [PDF]. SAS Support.
<https://support.sas.com/content/dam/SAS/support/en/products-solutions/ondemand/uploading-accessing-local-data.pdf>

Credé, M., & Kuncel, N. R. (2008). Study habits, skills, and attitudes: The third pillar supporting collegiate academic performance. *Perspectives on Psychological Science*, 3(6), 425–453. <https://journals.sagepub.com/doi/10.1111/j.1745-6924.2008.00089.x>

Curcio, G., Ferrara, M., & De Gennaro, L. (2006). Sleep loss, learning capacity and academic performance. *Sleep Medicine Reviews*, 10(5), 323–337.
<https://www.sciencedirect.com/science/article/pii/S1087079206000466>